

A half-plane radius for the small-measure Banaszczyk-transform question

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Source and question

The source paper is Maud Szusterman, *Monotonicity of the Gaussian measure under Banaszczyk transforms*, arXiv:2510.00842. It defines, for a closed convex set $K \subset \mathbb{R}^n$ and a nonzero vector u , a modified Banaszczyk transform

$$K \circ u = (K + [-u, u]) \cap (Z_K(u) + \mathbb{R}u),$$

where, putting $r = \|u\|_2$ and $I_y = (K - y) \cap \mathbb{R}u$,

$$Z_K(u) = \{y \in u^\perp : \gamma_1(I_y) \geq \gamma_1([-r, r])\}.$$

The paper proves monotonicity when $\gamma_n(K) \geq 1/2$ and $\|u\|_2 \leq 1/7$.

Question 4.1 of the paper asks whether there is an increasing function $r : (0, 1) \rightarrow (0, \infty)$ such that

$$\gamma_n(K) \geq p, \quad \|u\|_2 \leq r(p) \quad \implies \quad \gamma_n(K \circ u) \geq \gamma_n(K)$$

for all dimensions and convex bodies. The appendix explains that for $p < 1/2$ it is not clear whether the planar hypograph statement is equivalent to the weaker half-plane test:

$$\gamma_2(H) \geq p, \quad \|u\|_2 \leq r_0 \quad \implies \quad \gamma_2(H \circ u) \geq \gamma_2(H)$$

for all half-planes $H \subset \mathbb{R}^2$.

Result

Let

$$\phi(t) = \frac{e^{-t^2/2}}{\sqrt{2\pi}}, \quad \Phi(t) = \int_{-\infty}^t \phi(x) dx, \quad T(A) = \Phi(-A).$$

For $0 < r < 1$ define $w_r > 0$ by

$$T(w_r) = \Phi(-w_r) = \gamma_1([-r, r]) = 2\Phi(r) - 1,$$

and put $W_r = w_r - r$.

Theorem 1 (Half-plane small-measure radius). *Fix $0 < p < 1$ and set*

$$A_p = \max\{0, -\Phi^{-1}(p)\}.$$

There is $\rho_p > 0$ such that for every half-plane $H \subset \mathbb{R}^2$ with $\gamma_2(H) \geq p$ and every $u \in \mathbb{R}^2$ with $\|u\|_2 \leq \rho_p$,

$$\gamma_2(H \circ u) \geq \gamma_2(H).$$

More quantitatively, the conclusion holds for every $r = \|u\|_2 > 0$ satisfying

$$W_r > A_p \quad \text{and} \quad \frac{T\left(\sqrt{W_r^2 - A_p^2}\right)}{W_r} \leq r e^{-A_p r}. \quad (1)$$

Proof

By rotational invariance of Gaussian measure and the identity $K \circ u = K \circ (-u)$, it is enough to take $u = r e_2$ and to write the half-plane as

$$H = H(d, \alpha) = \{(x, y) : cx + sy \leq d\}, \quad c = \cos \alpha, \quad s = \sin \alpha, \quad \alpha \in [0, \pi/2].$$

Since $\gamma_2(H) = \Phi(d) \geq p$, we have

$$d \geq a_p := \Phi^{-1}(p) \geq -A_p.$$

The endpoint cases are immediate. If $s = 0$, then H is invariant under vertical enlargement and $H \circ r e_2 = H$. If $c = 0$, then $H = \{y \leq d\}$; once $W_r > A_p$ we have $\gamma_1([-r, r]) = \Phi(-w_r) \leq p \leq \Phi(d)$, so every vertical section is kept and $H \circ r e_2 = H + r e_2$.

Assume now that $0 < s, c < 1$. The vertical section of H over x is $(-\infty, (d - cx)/s]$, so the condition defining $Z_H(r e_2)$ is

$$\frac{d - cx}{s} \geq -w_r \quad \Longleftrightarrow \quad x \leq x_0 := \frac{d + s w_r}{c}.$$

Putting $D = d + sr$, the transform is the translated half-plane $H + r e_2 = \{cx + sy \leq D\}$ cut by $x \leq x_0$. Hence

$$\gamma_2(H \circ r e_2) - \gamma_2(H) = [\Phi(D) - \Phi(d)] - \gamma_2(C_r),$$

where

$$C_r = \{(x, y) : x \geq x_0, y \leq (D - cx)/s\}.$$

First estimate the gain from translating the half-plane. For $0 \leq t \leq sr$,

$$\frac{\phi(d + t)}{\phi(D)} = \exp\left(\frac{D^2 - (d + t)^2}{2}\right) = \exp\left(\frac{(sr - t)(D + d + t)}{2}\right) \geq e^{-A_p r},$$

because $sr - t \leq r$ and $D + d + t \geq 2d \geq -2A_p$. Therefore

$$\Phi(D) - \Phi(d) = \int_0^{sr} \phi(d + t) dt \geq s r e^{-A_p r} \phi(D). \quad (2)$$

For the cone, change variables $y = (cx - D)/s$. Since $cx_0 - D = s(w_r - r) = sW_r$,

$$\gamma_2(C_r) = \frac{s}{c} \int_{W_r}^{\infty} T(y) \phi\left(\frac{D + sy}{c}\right) dy.$$

Using Mills' inequality $T(y) \leq \phi(y)/y$ for $y > 0$, and $y \geq W_r$, gives

$$\gamma_2(C_r) \leq \frac{s}{cW_r} \int_{W_r}^{\infty} \phi(y) \phi\left(\frac{D+sy}{c}\right) dy.$$

The elementary identity

$$y^2 + \left(\frac{D+sy}{c}\right)^2 = D^2 + \left(\frac{y+sD}{c}\right)^2$$

then yields

$$\gamma_2(C_r) \leq \frac{s\phi(D)}{W_r} T\left(\frac{W_r+sD}{c}\right). \quad (3)$$

Since $D \geq -A_p$,

$$\frac{W_r+sD}{c} \geq \frac{W_r-sA_p}{\sqrt{1-s^2}}.$$

For $W_r > A_p$, the minimum of the right-hand side over $0 \leq s < 1$ is $\sqrt{W_r^2 - A_p^2}$. Thus

$$\gamma_2(C_r) \leq \frac{s\phi(D)}{W_r} T\left(\sqrt{W_r^2 - A_p^2}\right). \quad (4)$$

Combining (2) and (4), condition (1) implies

$$\gamma_2(H \circ re_2) - \gamma_2(H) \geq 0.$$

It remains only to see that (1) holds for all sufficiently small r . As $r \downarrow 0$,

$$T(w_r) = 2\Phi(r) - 1 \sim \sqrt{\frac{2}{\pi}} r, \quad w_r \rightarrow \infty, \quad rw_r \rightarrow 0.$$

Let $z_r = \sqrt{W_r^2 - A_p^2}$. Mills' asymptotic gives

$$\frac{T(z_r)}{T(w_r)} \sim \frac{w_r}{z_r} \exp\left(\frac{w_r^2 - z_r^2}{2}\right) \rightarrow e^{A_p^2/2}.$$

Consequently

$$\frac{T(z_r)}{W_r r} = \frac{T(z_r)}{T(w_r)} \frac{T(w_r)}{r} \frac{1}{W_r} \rightarrow 0,$$

whereas $e^{-A_p r} \rightarrow 1$. Thus (1) holds for every $0 < r \leq \rho_p$ after reducing $\rho_p > 0$ if necessary. This proves the theorem.

Limitations and review notes

The theorem proves only the half-plane test statement isolated in the appendix of the source paper. The reduction from arbitrary convex bodies of measure $p < 1/2$ to half-planes is explicitly identified there as unclear; this packet does not close that gap. The bound supplied by (1) is also only a sufficient lower-radius condition and is not asserted to match the almost-vertical half-plane upper bound mentioned by Szusterman.

The proof is analytic and uses only the source paper's definition of $K \circ u$, the exact two-dimensional half-plane formula, and the standard Mills tail inequality/asymptotic for the Gaussian distribution.

References

- Maud Szusterman, *Monotonicity of the Gaussian measure under Banaszczyk transforms*, arXiv:2510.00842.
- W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. I, for the standard Mills ratio estimates for Gaussian tails.